Investor Attention and Sentiment

Kamran Quddus[1] and Ashok Banerjee[2,3]

Abstract

This paper finds empirical support to the notion that non-market distraction events impair the ability of investors to incorporate firm relevant information into prices. We classify the distraction events into subcategories using machine learning algorithm. Second, we highlight that investors do not react homogeneously to the different non-market distraction events. For instance, a distraction induced by a political situation will produce smaller underreaction, whereas a distraction triggered by sports & entertainment events will induce larger underreaction. The empirical evidence is robust across models even after controlling for relevance, novelty, and sentiment of the news. The trading pattern of market participants suggests a significant decline in the number of transactions during various distraction periods indicating that investor inattention inhibits the decision making of investors. One important managerial implication of our results is that managers should endeavor to release any negative firm-specific news during such periods to avoid major price erosion in stock prices.

1. Introduction

Traditional asset pricing models suggest that stock prices quickly reflect all relevant information without delay. However, one cohort of literature has surprised the academia with empirical evidence of a delay in the incorporation of firm-specific information into security prices. They posit underreaction as a natural antecedent to this behavior. Many classical, as well as behavioral researchers, have inquisitively looked into the subject of underreaction. Cognitive sciences literature highlights that investor inattention may be a source of this underreaction (Loh, 2010).

[1] Doctoral Student, Indian Institute of Management Calcutta, Diamond Harbour Road, Joka, Kolkata -700104
E-mail address: kamranq13@iimcal.ac.in

[2] Professor (Finance), Indian Institute of Management Calcutta, Diamond Harbour Road, Joka, Kolkata -700104
E-mail address: ashok@iimcal.ac.in

[3] We thank Calcutta Stock Exchange IPF and Thomson Reuters for their support and funding

Researchers look at aggregate market outcomes such as extreme returns (Barber & Odean, 2008), trading volume (Hou, Xiong, & Peng, 2009), and media coverage (Da, Engelberg, & Gao, 2011) to judge whether investors are attentive to the stock. They base their theoretical understanding on the assumption that certain observable and latent market outcomes can influence the level of engagement of investors. For example, higher coverage of a particular stock in media may draw the attention of more investors, and that may lead to unusually high trading volume in the particular stock. Extending this line of thinking, we examine the market dynamics during certain fleeting periods marked by the systemic noise of a kind. For academic rigor, we name such systemic periods as distraction days and describe its characteristics in subsequent paragraphs.

Psychologists argue that limited cognitive resource constrains human thinking capacity. This constraint leads to continuous tension among the multiple information channels competing for limited mental resources (Egeth & Kahneman, 1975; Hirshleifer & Teoh, 2003). The cognitive model has aided the understanding of complex human interactions and has proved to be the quintessential missing link so far. Behavioral finance researchers have recently started exploring this line of thinking. This strand of research leans back on the theoretical underpinnings of cognitive limitation to justify the time delay in market reaction to firm related information.

Our aim in this study is to present new evidence on how competing stimuli impact investors' decision making. Although researchers argue that irrelevant stimuli can distract investors, few studies examine the impact of value irrelevant competing stimuli on decision making (Drake, Gee, & Thornock, 2016). We use distraction events covered on the front page of a newspaper as a proxy for investor inattention. We argue that these events act as a shock to retail attention that diverts the attention of investors away from the market. This momentary disturbance hinders their ability to quickly and comprehensively react to firm-specific information. We extend the investor attention literature by investigating the distraction effect using a list of key non-market distraction events that are for the most part value irrelevant. Our study brings two strands of literature that investigates the behavior of investors, the consumer of news (Dellavigna & Pollet, 2009) and the other that examines the role of media, the intermediary of news (Bagnoli, Clement, & Watts, 2005; Boulland, Degeorge, & Ginglinger, 2016). This study also addresses the concern as to whether some forms of distraction induces larger underreaction compared to others.

To illustrate investor inattention, we highlight one major political event that attracted nationwide attention as evident from national and local media coverage. In April 2011, Anna Hazare, a social activist launched India Against Corruption (IAC) campaign demanding stronger Lokpal (Ombudsman) Bill in India. The general public in India was keenly following the details of the incident and every major sequence of developments as it unfolded. Even though Anna and his team protested in Delhi, almost entire nation eagerly followed the developments through media. News channels witnessed a drastic jump in viewership with approximately 2.5 million new viewers joining every week during Hazare agitation compared to the previous week (Stancati & Pokharel, 2011). The IAC campaign was a media hit with a viewership of news channels jumping up by almost six percent[4] in the week through 27th August 2011. The incidence of coverage of related news in various media shot up nearly 1,000 percent in the same week[5].

What is more intriguing is the response of investors to corporate announcements during this period. As anecdotal evidence, we handpick a few corporate announcements (see Table I) during this time of distraction to convey our message. We collect news releases for six companies both inside and outside of the distraction period. We find surprising results. The three stocks viz. Infosys (NSE: INFY), Motherson Sumi (NSE: MOTHERSUMI) and Fortis Healthcare (NSE: FORTIS) showed a muted response to the firm-specific announcements within the distraction period. In contrast, a similar announcement outside the distraction period elicited smaller underreaction. One can always argue that the two news may have different material content and is expected to be received differently by the market. Hence we pictorially represent results for six firms spread across sectors. The results hint at underreaction to company-specific announcements during distraction days.

Table I
List of Company specific announcements

Company	Normal (N) / Distraction (D)	Announcement
Coal India	**N**	Coal India arms may be listed over next five years
	D	Coal India given 'maharatna' status; Neyveli Lignite now 'navratna'
Infosys	**N**	Infosys BPO's US arm gets multi-year order from Canada's FaithLife

[4] As per News Content Track, a tool of media research firm TAM. Refer The Wall Street Journal blog at http://blogs.wsj.com/indiarealtime/2011/09/05/why-was-hazare-such-a-media-hit/

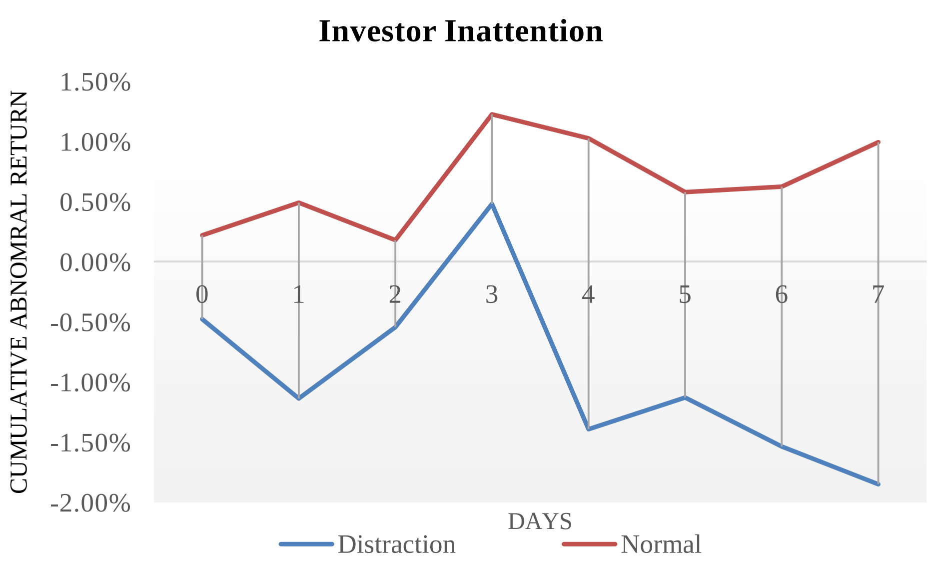
[5] Factiva is a global news database featuring nearly 33,000 news sources across geographies, including The Wall Street Journal, Dow Jones Newswires and Barron's

	D	Officials reportedly in talks to have Infosys set up local presence
Motherson Sumi	**N**	Motherson group to buy 80% in Germany's auto part co Peguform
	D	Delhi HC Approves Merger of Subsidiaries with Motherson Sumi
Oracle Financial Services	**N**	Oracle Financial gets IT contract from Tanzania's Amana Bank
	D	Bank of the Lao selects Oracle FLEXCUBE Universal Banking to Modernize Operations
Fortis Healthcare	**N**	Fortis Global Health to invest A$100 mln in Australia's Dental Corp
	D	India's Fortis Buying Way Into Asia-Pacific Health Care Market
Hexaware Tech	**N**	Hexaware Signs its largest deal to-date Potential revenue worth $ 177m over five years
	D	Hexaware wins multi-million dollar deal

We show the cumulative abnormal returns for firms during the distraction period as compared to the normal days (see Figure I). The figure demonstrates two interesting features ––

a) Investors' reaction is negative to firm-specific news during distraction periods, and
b) Prolonged period of distraction causes more damage to stock prices

Figure I
Investors underreact to firm-specific announcements during distraction periods
Cumulative abnormal return (CAR) = $\sum_{t=-2}^{n}$ AAR , where n = 0, 1,…7 days



The majority of the existing studies consider the impact of the distraction events on the pricing of earnings (Drake et al., 2016). We argue that earnings information is a backward looking number which appears with a lag. Investors react to all firm-related information promptly and do not wait for the reporting of the accounting number. Therefore we take a slightly different approach and consider the impact of a distraction event on the market response to all firm relevant information. This idea may not have been explored earlier for lack of authentic information on all firm relevant news in a structured format. We rely on Thomson Reuters News Analytics (TRNA) for their database on corporate news and sentiment scores. We briefly describe the TRNA database in a later section.

We make two major contributions to the behavioral finance literature. *First*, we find empirical support to the notion that non-market distraction events impair the ability of investors to incorporate relevant information into prices. We classify the distraction events into subcategories using machine learning algorithm. *Second*, we highlight that investors do not react homogeneously to the different non-market distraction events. For instance, a distraction by natural calamities and disasters will produce smaller underreaction, whereas a distraction triggered by law and order situation will have larger underreaction. The empirical evidence is robust across models even after controlling for relevance, novelty, and sentiment of the news.

Cognitive theorists suggest that human being have a limited mental capacity. Psychologists indicate that when individuals are subjected to multiple stimuli, the information sources stack up in a queue and get priority based on which data source is more pressing. This constraint results in selective allocation of time and attention (Hirshleifer & Teoh, 2003). Also, other considerations such as complexity of the task, intelligence, and tiredness of brain dictate the whole dynamics. There has been a growing body of research investigating the impact of limited cognitive capability on investor decision-making (Lim and Teoh 2010).

Contemporary studies, which have examined the impact of investor attention, look at anticipated events such as NCAA basketball tournament (Drake et al., 2016). Our work differs from previous studies in that we examine major systemic distractions which were largely unexpected such as a major earthquake, political crisis, among others. Moreover, we filter out these systemic events through a rigorous and scientific approach. These non-market distraction events concern a vast majority of the nation as demonstrated through their coverage in the popular media. Earlier studies hint at the possible interaction of distraction with investor biases, such as “mood swings” triggered by World Cup soccer games impacting the stock market outcomes (Edmans, Garcia, &

Norli, 2007). We allow this potential interaction of investor biases with a variety of distraction events.

Several studies demonstrate market underreaction by using proxies for investor inattention such as trading volume (Hou et al., 2009), event occurrence on Fridays (Dellavigna & Pollet, 2009), down market periods (Hou et al., 2009), and non-trading hours (Francis, Pagach, & Stephan, 1992). Barber & Odean (2008) focus on the buying behavior of investors in attention-grabbing stocks. They hypothesize that investors face a search cost while choosing from a large subset of stocks and that stocks in the news, stocks experiencing high abnormal trading volume and stocks with an extreme one-day returns draw their attention and form the subset to choose from. (Odean, 1999) suggests that investors choose from attention-grabbing stocks based on their preferences.

The rest of the paper is organized as follows. Section 2 provides a description of the data and summary statistics. Section 3 presents hypothesis development and model. Section 4 provides the detailed results on distraction effect. Section 5 provides additional robustness tests. Section 6 concludes the paper.

2. Data

Our sample consists of all firms listed on the National Stock Exchange of India (NSE). We collect the adjusted stock price data for 1,789 Indian companies from the Centre for Monitoring Indian Economy (CMIE) Prowess database. The database assigns a unique CMIE company code to each firm, which acts as a unique identity for the enterprise.

We avail the Times of India (TOI)[6] archives for the event headlines data to prepare the distraction events database. The Google Trends makes available the Search Volume Index (SVI) scores which help in gauging the general interest level of individuals for the events. Further, we also collect the frequency count of the number of times a news event gets reported in other popular media sources. We use this frequency count to filter out the regional or less popular events from our database. Finally, NSE’s proprietary tick-by-tick data helps our analysis by focussing on buying or selling patterns of different categories of traders. The database has

---

[6] TOI is is the largest selling English language daily in the world (Audit Bureau of Circulations, 2015) and has been ranked among the world’s six best newspapers (BBC, 1991)

separate flags to indicate whether the trade was initiated by an algorithmic or a non-algorithmic trader. The client's field of the database is particularly useful in identifying the customer type on each side of the trade.

The majority of the studies in the past consider indirect proxies for investor attention. These proxies are mostly endogenous and noisy which may affect the reliability of the results. We propose that external non-market events that are value irrelevant may act as better proxies for investor distraction. Irrelevant stimuli are distracting, echo the core essence of limited attention models deciphering the market underreaction anomalies (Hirshleifer, Lim, & Teoh, 2009). Our empirical settings offer an additional benefit that unlike seasonal earnings announcements, firm-specific announcements occur throughout the year. Further our choice of settings allows clearly defined categories of investor distraction that also captures the interaction of investor biases and cognitive attention.

We use the historical tick-by-tick data for sample firms from NSE for investigating the trading pattern of the different categories of traders. The data gives the record of all trades executed on NSE along with a timestamp and other particulars of trade, such as the number of shares bought or sold, average transaction price, stock code, order type and other metadata. The high-frequency NSE proprietary data has an algo indicator field that takes values 0,1,2 and 3. The algo indicator values 0 and 2[7] indicate trades are initiated by members that are using the algorithmic trading facility, whereas values 1 and 3[8] identify trades executed by non-algorithmic terminals. Similarly, client identity flag takes values 1, 2 and 3 for proprietary, client and custodian trades respectively.

2.1 Identification of Distraction Events

We follow a novel approach to collect a list of irrelevant distraction events. We begin by scanning the news headlines and bylines of the TOI archives. Selecting a media source such as TOI does not induce any biases as the speed of news dissemination does not drive investor attention (Boulland et al., 2016). We look up the keywords appearing in the TOI headlines on Google Trends to examine the nationwide relative popularity of the search term. Google Trends

[7] The algo indicator value two indicates trades are initiated by algorithmic traders using Smart Order Routing (SOR)
[8] The algo indicator value three indicates trades are initiated by non-algorithmic traders using Smart Order Routing (SOR)

disseminate the Search Volume Index (SVI) scores that facilitate easier comparison across terms for its search popularity on Google Search. The numbers are scaled to a range of zero to hundred. A larger score indicates higher popularity of the phrase. This helps in assessing the general level of interest for a particular news and gives a reasonable indication that individuals were actively searching for that news event. The choice of SVI is also motivated by (Da et al., 2011) who argue that the score captures investor attention in a more timely fashion and represents a direct and unambiguous measure of attention.

We collect the news headlines from the Times of India archives over a 12 year period from January 2004 through December 2015. The choice of the period was constrained by the availability of the records on the site. There were 49 holidays during the sample period when TOI was not issued. We examine headlines and bylines on the front page from the 4,334 available TOI editions. Since our goal is to construct a database of the major distraction events, we filter out the news items where Google SVI index is less than 100. This gives us a list of 368 key distraction events during the sample period. We apply a second filter to check if these incidents were sufficiently covered in other news media as well. For this purpose, we remove the items which appeared for less than 50 times during the month in which the news made headlines. Using this procedure, we get a final list of 333 key distraction events.

Table II
Descriptive Statistics on Distraction Events

The figure gives the yearly distribution of the number of event days of each category of distraction events during the period 2004 – 2015. The news headlines have been collected from the Times of India archives and categorized using Non-negative matrix factorization algorithm

	Group				
	(1)	(2)	(3)	(4)	
Year	Natural Calamities & Disasters	Political	Law & Order	Sports & Entertainment	Total
2004	9	11	6	11	37
2005	5	3	3	4	15
2006	4	5	9	3	21
2007	3	7	4	11	25
2008	12	1	7	6	26
2009	23	11	6	13	53
2010	7	10	11	11	39
2011	5	8	6	5	24

2012	4	6	2	6	18
2013	1	4	6	8	19
2014	3	14	2	9	28
2015	6	9	3	10	28
Total	82	89	65	97	333

We start by breaking each sentence in the news articles into words, through a process called tokenization[9]. We subject the tokenized words through subsequent processing steps such as stemming and lemmatization. Stemming replaces each word with its root word, and lemmatization performs a morphological analysis of words to return words in its base form commonly referred to as lemma. One common goal of these preprocessing steps is to neutralize the jargons and acronyms and minimize the variation problems through a vocabulary control technique. The tokenized texts are assigned a part-of-speech tag using Stanford CoreNLP[10]. We also eliminate redundant entities such as proper nouns, dates, and digits as they do not convey extra information in classifying the text into different topics.

Researchers use linguistic-based analytical models to explore unstructured data in textual format. These analytical models have been useful in extracting information in non-conventional form from various channels and find useful application in social media analytics (Lim, Buntine, Chen, & Du, 2016), sentiment analysis (Liu, 2012; Pang & Lee, 2008) and topic modeling (Blei, 2012)

We use a nonparametric Bayesian model for eliciting the hidden themes in the text corpus. The basic structure resembles latent variable models. Studies in machine learning use probabilistic algorithms to unravel and annotate large archives of documents with thematic information (Blei, 2012).

Topic modeling relies on statistical techniques to examine the words of the original text to discern the major themes hidden in the document. The algorithm does not require any prior labeling of the documents. One of the primary aims of this technique is to examine topic coverage of the news text. We start with an n-gram based topic model to identify the specific

[9] Many text processing software have inbuilt functionality for text-processing. One such Python library is Natural Language ToolKit (NLTK) which is commonly used in text processing and analytics

[10] Stanford CoreNLP toolkit, is an extensible pipeline that provides core natural language analysis. This toolkit is quite widely used, both in the research NLP community and also among commercial and government users of open source NLP technology (Manning et al. 2014)

topic covered in the news. Finally, we use non-negative matrix factorization to elicit the topic coverage over time.

2.2 Non-negative Matrix Factorization

Text-based analytical models have found extended use in health diagnostic (Zhang et al., 2013), image processing (Xue, Tong, & Yuan, 2014), and text mining (Barman, Iqbal, & Lee, 2006). We use an unsupervised machine learning algorithm for text classification. Using unsupervised Non-negative Matrix Factorization (NMF) overcomes the limitations of prior labeling of training data (Feng, Jiao, Lv, & Zhou, 2016). In practice, we frequently come across data set with no apriori labeling.

The model assumes that k number of topics exists over the whole collection of documents. As represented in figure II, each topic β_k can be assumed to have a certain distribution over the words present in each document. We assume that i^{th} item in the corpus has a certain allocation of topics represented by θ_{ik}.

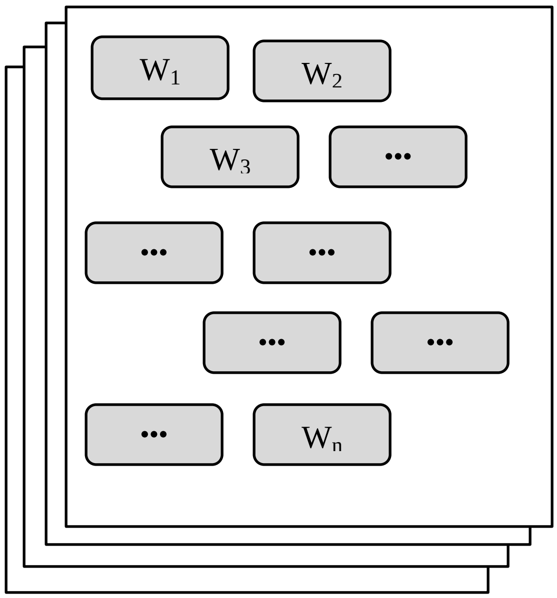
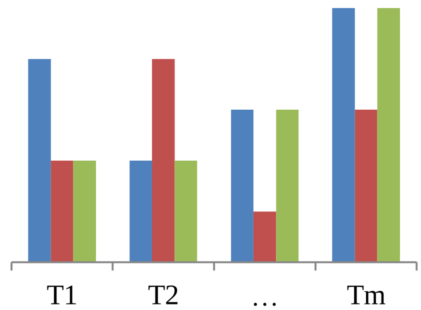
Linguistic researchers proceed by analyzing N-gram frequencies for deciphering information over a collection of texts. The algorithm arranges texts in a document-term matrix (DTM), and the proximity between any two news items is assessed by calculating the Euclidean distance between two pairs of word frequencies. Given two text items represented by vectors $\overrightarrow{z_1}$ and $\overrightarrow{z_2}$ in n-dimensional space, the Euclidean distance between the items is represented by the expression below:

$$\|\overrightarrow{z_1} - \overrightarrow{z_2}\| = \sqrt{\sum_{j=1}^{n} (z_1 - z_2)^2}$$

Since Euclidean distances represent the distance between two items, the inverse of the distance can be used as a measure of proximity.

Figure II:
Non-negative matrix factorization (NMF)

NMF assumes k number of topics exist for the entire corpus. Each of the k^{th} topics is a distribution of m keywords with probability p_{mi}. These themes are mapped onto the document to assess the presence of k topics. W_i's are the words present in the document.

Topics	Documents	Topic proportions and assignments
K_{11} p_{11} K_{12} p_{12} K_{13} p_{13} ••• K_{21} p_{21} K_{22} p_{22} K_{23} p_{23} ••• K_{m1} p_{m1} K_{m2} p_{m2} K_{m3} p_{m3} •••		

We focus on news archives from print media due to their wide circulation and content reliability. One principal argument for looking at the front page headline events is that only significant events appear in the headlines on the front page. Therefore these events concern large population of a nation and immediately catch their attention. Using a scientific processing technique described in the earlier paragraph, we divide the distraction events into four sub-groups, namely —– Natural calamities & disasters, political, law & order and sports & entertainment. The labeling of each of the sub-groups follows from the semi-supervised natural language processing algorithm. NMF method tries to form the distribution of top keywords appearing in each news text based on its probability of occurrence. As a consequence, a news text conforms to a certain sub-group based on the likelihood of occurrence of particular keywords from that defined group. For example, a higher occurrence of words such as "game," "cricket," "team," etc. shows that the news is more likely about a sports event. Table II provides year-wise, detailed breakup on each of the sub-categories. Sports & entertainment events form the largest category with a count of 97, followed by political (89), natural calamities (82) and law & order (65).

Once we obtain the distraction events, our next task is to get sentiment of firm-specific news. The sentiment scores for the firm-specific news are available from the Thomson Reuters News Analytics (TRNA). Using a sophisticated computational linguistic process, TRNA deciphers each news item and scores it for each asset appearing in the news. The primary attributes of the TRNA score are categorized as follows:

- Relevance: Gives a quantitative measure of how relevant the news item is to the firm mentioned in the news
- Sentiment: Indicates the tone of the news item
- Novelty: Shows whether the news item is unique or related to some previously seen news item
- Volume: Quantitative measurement of the number of times, the mentioned firm appeared in other news items in history
- Headline Classification: Gives brief analysis of the headline

We rely on TRNA for firm-specific news sample from 2004 to 2015. Our treatment sample consists of all firms for which sentiment scores and other metadata were available corresponding to press releases archived on Reuters Data Feed (RDF).

Table III shows descriptive statistics of the firm-specific sentiment scores for Indian firms from January 2004 till December 2015. A total of 990,003 news was released over the period. Panel A shows that on average 1.8 percent of the 1,789 Indian firms have some news available on TRNA. The daily news distribution has a standard deviation of 1.4. On the lower side of the distribution, there were days where only 0.076% of the total firms had some news releases. Whereas, on the higher end of the distribution, there were days where almost 5% of the firms had some announcements. The mean sentiment per news release is 0.091. Similarly, Panel B through D gives similar breakup by focussing on the predominant sentiment of the news. For example, Panel B gives the statistics on positive sentiment news.

Table III
Descriptive Statistics on Firm-Specific Sentiment Scores

	Mean	S.D.	1%	50%	99%
Panel A: All News Releases and sentiment (990,003 observations)					
News stocks per day (% of 1,789 firms)	1.863	1.406	0.072	1.627	4.706
News days per stock (% of 4,479 days)	0.235	0.014	0.200	0.236	0.255
Stocks per news release	4.523	5.960	2.000	2.000	30.000
Sentiment per news release	0.091	0.386	-0.763	0.093	0.820
Panel B: News releases with dominant Positive Sentiment (339,706 observations)					
News stocks per day (% of 1,780 firms)	1.977	1.387	0.076	1.788	4.959
News days per stock (% of 4,265 days)	0.245	0.015	0.210	0.245	0.269
Stocks per news release	4.212	4.944	2.000	2.000	26.000
Sentiment per news release	0.456	0.238	0.018	0.481	0.826
Panel C: News releases with dominant Negative Sentiment (248,166 observations)					
News stocks per day (% of 1,756 firms)	2.014	1.457	0.058	1.775	5.135
News days per stock (% of 4,182 days)	0.258	0.014	0.218	0.261	0.278
Stocks per news release	4.616	5.316	2.000	2.000	26.000
Sentiment per news release	-0.425	0.224	-0.764	-0.503	-0.014
Panel C: News releases with Neutral Sentiment (402,131 observations)					
News stocks per day (% of 1,782 firms)	2.068	1.683	0.080	1.759	5.198
News days per stock (% of 4,056 days)	0.262	0.014	0.225	0.264	0.283
Stocks per news release	2.755	2.743	2.000	2.000	16.000
Sentiment per news release	0.102	0.101	-0.134	0.077	0.445

Contemporary research in finance highlights the role of limited investor attention in asset pricing. (Dellavigna & Pollet, 2009) show that investors underreact to earnings announcement on Fridays compared to other weekdays. They predicate the findings on the assumption that investors are more likely to be inattentive on Fridays relative to other weekdays. Despite the widening appeal for cognitive factors such as limited attention playing a vital role in explaining asset pricing dynamics, there has been a scarcity of empirical research in this area (Dellavigna & Pollet, 2009).

Our study endeavors to contribute to the existing literature on stock market underreaction by investigating the inattention of investors and the manner in which it affects the decision making of investors. The idea resonates well with the hypothesis of investor attention impacting the quality of decision making (Dellavigna & Pollet, 2009), determining stock buying decision of retail investors (Barber & Odean, 2008), affecting the trading behavior of investors (Yuan, 2015) and explaining market reaction to macroeconomic news announcements (Chen, Liu, Lu, & Tang, 2016).

To further strengthen our point, we investigate the market response to company-specific news announcements in various distraction periods. We collect at firm related announcements from Dow Jones Factiva. Analysis of stock market reaction reveals that similar news announcements generate different market response during natural calamities and disaster situation as compared to the market reaction during a distraction triggered by an attention-grabbing law & order situation. This difference in stock return response cannot be ruled out as a non-significant aberration owing to the random walk movement of share prices. This differential response merits a detailed and more rigorous examination that can lead to more sophisticated explanations of existing phenomena, for example, underreaction to earnings announcements on Fridays (Dellavigna & Pollet, 2009). The difference in stock return response also hints at the possibility of nature of distraction itself affecting the decision-making behavior of investors.

Contemporary studies in finance look at both the indirect proxies of investor attention, such as trading volume (Barber & Odean, 2008), extreme returns (Barber & Odean, 2008), news headlines ((Barber & Odean (2008), (Yuan, 2015))[11] as well as direct proxies such as Google Search Frequency (Da, Engelberg, & Gao, 2011). Most of the previous studies just look at proxies of investor attention without examining the source of distraction itself. We contribute to the existing investor attention literature by taking a more granular look into the nature of the systemic distraction and the investor's response to such nation-wide shocks. We hypothesize that the nature of distraction itself can have implications on the way investors react to distractions. Previous research suggests that distraction effect of each category of distraction are distinct and interact with other investor behavioral biases such as investor mood (Drake et al., 2016). To the best of our knowledge, no other study has brought this distinction and showed this interaction in such a scientific setup.

[11] See (Da et al., 2011) for a more detailed discussion

3. Hypothesis and Research design

Earlier studies have looked into the effect of investor sentiment in stock prices (Baker & Wurgler, 2006) and into the issue of mispricing of securities (Baker & Wurgler, 2007). Drake et al. (2016) examine the effect of competing stimuli on markets during periods coinciding with the NCAA basketball tournament. The tournament is commonly referred to as March Madness as it commences in March and generates "extraordinary levels of excitement." They find that March Madness veers investors' attention away from new earnings announcements. Number of trades is found to be significantly lower for firms announcing their earnings during the tournament periods compared to nontournament periods.

We examine whether irrelevant stimuli triggered by key non-market distraction events, hinder market reaction to firm-specific announcements. Given the cognitive and temporal limits to the information processing capacity of individuals, we hypothesize that market reaction to any firm specific announcements during such distraction periods will elicit a muted response. Therefore we hypothesize that strength of the relationship between the sentiment of any firm specific announcement to the stock return will be less during periods marked by distraction compared to normal periods.

H1: News carrying positive or negative sentiment will elicit muted response during distraction periods relative to normal trading days

We break H1 into two parts and test it separately for news carrying positive and negative sentiment respectively. This is motivated by earlier behavioral studies that suggest that investors react differently to positive and negative announcements (Hirshleifer et al., 2009).

H1(a): News carrying positive sentiment will elicit muted response during distraction periods relative to normal trading days

H1(b): News carrying negative sentiment will elicit muted response during distraction periods relative to normal trading days

We also contemplate that all kinds of distraction will not trigger similar levels of underreaction. This is because different categories of distraction have distinctive inherent ability to divert the focus of individuals. This also allows us to delineate the interactions of investor biases with news sentiment clearly.

Our second hypothesis is motivated by studies that argue that investors react differently to the positive and negative news. For example, (Hirshleifer et al., 2009) indicate that investors respond differently to positive and negative earnings surprises by firms.

H2: Investors react differently to different categories of distraction

With advancements in technology, individuals have gradually transitioned towards the use of more algorithmic as opposed to manual trading strategies. We argue that the investor inattention hypothesis would not hold merit under such scenarios. Therefore, we use tick-level data from NSE to examine this issue in further detail. This tick-by-tick proprietary data flags the trades executed using machine algorithms vis-à-vis non-algorithmic trades.

De Long, Shleifer, Summers, & Waldmann (1990) posit that investors are vulnerable to sentiment, which reinforces their belief about future cash flows. Earlier behavioral finance studies highlight the distinctive trading behavior among different group of investors. For example, the limits to arbitrage circumvent the arbitrageurs to trade on any market mispricing aggressively (Shleifer & Vishny, 1997). Thus the rational traders are less sanguine in acting on any sentimental news. Baker & Wurgler (2007) maintain that researchers now face the issue of measuring and quantifying the effects of sentiment and exploring the mechanism through which attentional constraints affects investors trading activity. Chakrabarty & Moulton (2012) find that moving towards a more automated business environment results in attenuating the effect of attentional limitations.

H3: Non-Algorithmic traders are more susceptible to extraneous distractions compared to Algorithmic traders

We take the “bottom up” approach suggested in (Baker & Wurgler, 2007) by highlighting the individual differences in biases among the different group of investors. Various psychological biases such as conservatism (Barberis, Shleifer, & Vishny, 1998), representativeness, and overconfidence (Daniel, Hirshleifer, & Subrahmanyam, 1998) may induce predictive ability in explaining the differences in underreaction or overreaction exhibited by investors.

We bring out the differentiation that even within non-algorithmic trades, less sophisticated traders would be more inattentive compared to institutional traders. Our proposition tests the psychological biases arising from the differences of opinion (Miller, 1977) among various categories of investors. The behavioral finance literature model investors as either rational arbitrageur who are less susceptible to changes in sentiment or the irrational traders, who are more prone to fluctuating sentiments (De Long et al., 1990).

H4: Less sophisticated (retail) traders are more affected by distractions as compared to institutional investors

We begin our analysis by collecting a list of distraction events using the TOI archives. These distraction events have a very high search interest as indicated by the Google SVI scores hitting 100 around these developments. Different media sources have well covered these distraction events. This is evident from the number of times these events appeared in headlines of other well-circulated news sources. Factiva aggregates this information from the all domestic and global news sources. Using the above-discussed filters, we generate a list of 333 key distraction events that act as a proxy to investor inattention. The intersection of distraction days with the TRNA news announcements days gives us a final sample of 26,065 news announcements on distraction days.

We categorize the events into four sub-categories using an unsupervised machine learning algorithm, viz NMF technique. The rationale for using this method is to eliminate the bias that would have crept if the distribution would have been done manually. Using a scientific machine learning algorithm ensures little room for individual subjectivity.

The company-specific announcements are available from TRNA. Each news item comes with a timestamp and Reuters Instrument Code (RIC) along with the sentiment field and other news metadata. A categorical variable that takes values +1, 0 and -1 respectively for positive, neutral and negative sentiment indicates the dominant sentiment class. The probability of each news belonging to each of the earlier mentioned categories is also provided by separate continuous variables for each sentiment class.

To investigate the investor behavior during distraction periods, we investigate the abnormal stock return around the news announcement dates. We use the distraction events as a proxy for investor inattention. Investor inattention hypothesis argues that these developments inhibit the ability of market participants to react to the firm-specific news. We use the model as shown in equation (1) to estimate the stock returns. In the first stage, stock returns are regressed on its one period lagged return, market return and a set of variables to control for the day of the week effect and non-weekend effect.

$$R_{it} = \beta_0 + \beta_{1i} R_{it-1} + \beta_{2i} R_{Mt} + \beta_{3i} D_t + \beta_{4i} Q_t + \varepsilon_{it}, \qquad \ldots\ldots..(1)$$

$D_t = \{ D_{1t}, D_{2t}, D_{3t}, D_{4t} \}$ are dummy variables for Monday through Thursday,

$Q_t = \{Q_{1t}, Q_{2t}, Q_{3t}, Q_{4t}, Q_{5t} \}$ are dummy variables for days for which previous 1 through 5 days are non-weekend holidays

$$\widehat{\epsilon_{it}} = \gamma_0 + \gamma_1 \text{ sent\_pos}_{it} + \gamma_2 \text{ sent\_neg}_{it} + \gamma_3 \text{ relevance}_{it} + \gamma_4 \text{ novelty}_{it} + \gamma_5 \text{ size}_{it} + (\text{Industry Dummies})_i + (\text{Year Dummies})_t + v_i$$

$$\ldots\ldots\ldots\ldots\ldots.(2)$$

$\widehat{\epsilon_{it}}$ are the residuals derived from previous regression, $sent\_pos_{it}$ and $sent\_neg_{it}$ are probability that the sentiment of the news was positive and negative respectively; $relevance_{it}$ measures the pertinence of the asset reported in the news; novelty is the measure of uniqueness of the news being reported

In the second stage, we take the residuals from regression (1) and regress it on sentiment scores of news releases. Positive and negative sentiment scores are represented by sent_pos and sent_neg respectively. The sentiment scores are scaled on a scale of zero to one, and hence can also be interpreted as the probability of news carrying positive and negative sentiment respectively. Hirshleifer et al. (2009) argue that smaller firms exhibit larger underreaction during distraction phases. We control for the size of the business to check for this variation in distraction effect.

4. Results

Table IV presents our empirical findings exhibiting the asymmetric response of investors to firm-specific announcements during distraction periods. Panel A shows the response coefficients to positive and negative sentiment news during all distraction periods. Overall the results indicate that the investors are distracted by irrelevant stimuli. The beta coefficient, γ_2 of -0.017 ($p > 0.10$), suggests that investors underreact to any negative corporate announcements during distraction phases. Surprisingly, γ_1 signifying the response coefficient for positive news is 0.40 ($p < 0.01$) is statistically significant. Panels B through E present the results across various distraction categories. We find empirical evidence of underreaction to negative sentiment news during all distraction periods, consistent with H1(b). However, our results statistically reject H1(a) as the coefficient γ_1 is economically and statistically significant in the overall distraction period as well as in different categories of distractions, except political events (Panel C). The γ_1 coefficients also vary from 0.4 ($p < 0.10$) in Panel B to 0.507 ($p < 0.05$) in Panel D, validating our H2. In other words, the results indicate that investors' response to a positive sentiment news differs across various categories of distraction. The relevance of news announcement has no statistical significance across distraction categories, excepting Panel B, where γ_3 is -0.290 ($p < 0.10$). This supports the investor inattention hypothesis that investors fail to react to any relevant news during distraction periods adequately.

Table IV

Asymmetric investor reaction to firm specific announcements during various distraction events

$$R_{it} = \beta_0 + \beta_{1i} R_{it-1} + \beta_{2i} R_{Mt} + \beta_{3i} D_t + \beta_{4i} Q_t + \varepsilon_{it},$$

$D_t = \{ D_{1t}, D_{2t}, D_{3t}, D_{4t} \}$ are dummy variables for Monday through Thursday,

$Q_t = \{ Q_{1t}, Q_{2t}, Q_{3t}, Q_{4t}, Q_{5t} \}$ are dummy variables for days for which previous 1 through 5 days are non-weekend holidays

$$\widehat{\epsilon_{it}} = \gamma_0 + \gamma_1 \, sent\_pos_{it} + \gamma_2 \, sent\_neg_{it} + \gamma_3 \, relevance_{it} + \gamma_4 \, novelty_{it} + \gamma_5 \, size_{it} + (\text{Industry Dummies})_i + (\text{Year Dummies})_t + v_i$$

$\widehat{\epsilon_{it}}$ are the residuals derived from previous regression, $sent\_pos_{it}$ and $sent\_neg_{it}$ are probability that the sentiment of the news was positive and negative respectively; $relevance_{it}$ measures the pertinence of the asset reported in the news; novelty is the measure of uniqueness of the news being reported

	Panel A		Panel B		Panel C		Panel D		Panel E	
	All Distraction Days		**Natural Calamities & Disasters**		**Political**		**Law & Order**		**Sports & Entertainment**	
	Coef.	Robust Std. Error	Coef.	Robust Std. Error	Coef.	Robust Std. Error	Coef.	Robust Std. Error	Coef.	Robust Std. Error
Sent_pos	0.400	0.111***	0.400	0.237*	0.136	0.209	0.507	0.213**	0.422	0.193**
Sent_neg	-0.017	0.098	-0.036	0.192	-0.125	0.174	0.166	0.195	-0.116	0.213
Relevance	-0.098	0.071	-0.290	0.146*	-0.100	0.128	0.050	0.142	-0.111	0.142
Novelty	0.003	0.010	-0.013	0.029	0.029	0.015*	-0.007	0.024	-0.005	0.013
Size	-0.029	0.012**	-0.075	0.027***	-0.015	0.019	-0.002	0.019	-0.024	0.023
_Cons	0.543	0.352	1.170	0.512	1.062	0.778	-0.718	0.469	0.109	0.325
Industry Dummies	Yes		Yes		Yes		Yes		Yes	
Year Dummies	Yes		Yes		Yes		Yes		Yes	
N	26,065		6,380		7,152		5,330		7,203	

Notes: Industries are defined by the Fama-French 48-industry classification. Variables are winsorized at the 1 percent and 99 percent levels. Standard errors are clustered by the news announcement date. *,**, and *** indicate significance at the $p < 0.10$, $p < 0.05$, and $p < 0.01$ levels respectively

5. Robustness tests and additional analyses

We provide additional tests to validate our hypothesis of investors getting distracted by external irrelevant stimuli. Table V presents empirical evidence on whether investors' reaction to news announcements during distraction periods remain muted. The intraday trading records are taken from NSE proprietary tick-by-tick data. Panel A shows the trading behavior of investors corresponding to news releases with positive sentiment. The number of shares traded and turnover are cumulated across various distraction periods and then averaged across the number of firms. Transactions give the average count of the number of trades executed.

Table V

Comparison of Market Activity during various distraction periods

Quantity traded refers to the average number of shares traded, Turnover measures the average quantity of shares traded (INR million), and Transactions quantifies the mean number of trades

Panel A: Volume Analysis for news with positive sentiment			
	Quantity Traded	Turnover	Transactions
Natural Calamities & Disasters	1,802,541	508	16,627
Political	2,142,673	581	18,670
Law & Order	1,437,856	492	15,650
Sports & Entertainment	1,785,927	494	16,362
All Distraction Days	1,810,060	520	16,888
Non Distraction Days	2,414,351	731	17,836
Difference	-604,291 [t = -6.36]***	-211 [t = -10.26]***	-948 [t = -2.11]***
Panel B: Volume Analysis for news with negative sentiment			
	Quantity Traded	Turnover	Transactions
Natural Calamities & Disasters	2,156,518	658	20,390
Political	2,426,712	719	21,861
Law & Order	2,040,093	631	19,886
Sports & Entertainment	2,409,612	651	21,172
All Distraction Days	2,280,051	668	20,926
Non Distraction Days	2,431,057	868	19,950
Difference	-151,006 [t = -1.2838]	-200 [t = -6.81]***	976 [t=1.5993]

Abnormal returns may not always capture the level of distraction—the trading activities do. Therefore, we examine the market activity of investors during various periods. Panel A shows that trading activity decreases during distraction periods compared to normal days. The difference in turnover is -211 ($p < 0.01$) and is statistically significant. Turnover and number of transactions are lowest during distraction phases caused by sports and entertainment events. Our empirical evidence suggests that investors are least distracted during political events. The turnover and number of transactions are highest during this period. Panel B shows trading activity for news with negative sentiment. Consistent with H2, investors also underreact to negative news during various transaction periods. The difference in turnover is -200 ($p < 0.01$) and is highly significant.

Our empirical analysis in Table IV and Table V demonstrate that investors may underreact to both positive and negative sentiment news during periods of distraction. Psychological theories document that individuals face a cognitive constraint that hinders their brain processing capacity. This may be true for less sophisticated traders— most notably retail traders. However, institutional traders are more likely to engage highly sophisticated computer algorithms and therefore are less susceptible to attentional constraints (Barber & Odean, 2008). This raises questions about the effect of attention constraints on price efficiency. Chakrabarty, Moulton, and Wang (2015) find that high-frequency traders (HFTs) moderate the underreaction due to attentional limitations.

We further analyze trading behavior according to client type (hypothesis H3). Table VI gives a breakup of trading activity by client type. We use the trade records from the NSE's tick-by-tick proprietary data in the cash market segment. The algorithmic indicator in the database allows us to identify the trades executed by algorithmic terminals as well as the non-algorithmic facility[12]. Further, we use client identity flag to segregate trading data based on whether the trade is proprietary, done on behalf of a client, or non-client non-proprietary trade. Panel A shows the trading activity by trader type for positive sentiment news. Consistent with H3, we find statistical evidence that inattention effect is more pervasive for non-algorithmic trades compared to algorithmic trades. The difference in trading volume between distraction days and normal days is

[12] The Algo indicator takes values zero through three for Algo, Non-Algo, Algo through SOR, and Non-Algo through SOR respectively. We combine the categories zero and two to represent the algorithmic trades. Similarly we combine the categories one and three to represent non-algorithmic trades.Trades through Smart Order Routing (SOR) represent a small fraction in both algo and non-algo transactions.

1,487 (p > 0.9) for algorithmic trading on behalf of the client. The corresponding difference is -3,625 (p < 0.01) for non-algorithmic client trades. Therefore, the non-algorithmic traders' level of activity in the market went down significantly during distraction periods — such trades were executed by humans.

We then check how the behavior of individual investors varies across different distraction periods. In particular, we check whether the retail ownership in a particular stock is significantly correlated with the cumulative abnormal returns. Retail_pct indicates the percent holding by retail investors in a specific stock. The results indicated in Table VII shows that retail investors are distracted during periods of distraction. This applies to both positive and negative sentiment news. This is consistent with H4 that retail investors are more distracted by external distractions than sophisticated institutional investors.

Table VI

Comparison of Market Activity by Trader Type

The figures indicate traded volume (INR million) by various categories of traders. The trading records were obtained using NSE tick-by-tick proprietary data and aggregated across various distraction days. The client identity was flagged using an indicator variable

Panel A: Market Activity by Trader Type for news with positive sentiment

	Algorithmic			Non-Algorithmic		
	Client	Proprietary	Non-CP Non-Prop	Client	Proprietary	Non-CP Non-Prop
	(1)	(2)	(3)	(4)	(5)	(6)
Natural Calamities & Disaster	16,246	11,125	4,125	12,769	10,594	35,829
Political	38,162	27,205	9,827	22,580	16,267	53,237
Law & Order	16,780	10,220	3,939	13,134	8,037	29,678
Sports & Entertainment	34,742	18,579	8,480	18,921	11,244	43,485
All Distraction Days	35,305	21,366	8,849	21,703	14,229	52,582
Non Distraction Days	33,818	17,169	7,860	25,328	14,502	54,211
Difference	1,487	4,197	989	-3,625	-273	-1,629
p-val	0.909	0.999	0.999	0.003	0.274	0.228

Panel B: Market Activity by Trader Type for news with negative sentiment						
	Algorithmic			Non-Algorithmic		
	Client	Proprietary	Non-CP Non-Prop	Client	Proprietary	Non-CP Non-Prop
	(1)	(2)	(3)	(1)	(2)	(3)
Natural Calamities & Disaster	24,338	14,387	5,398	18,689	12,736	43,724
Political	43,501	25,634	9,678	26,893	15,050	56,839
Law & Order	23,377	13,143	4,728	17,076	9,860	34,903
Sports & Entertainment	42,683	22,842	9,665	23,734	13,794	51,340
All Distraction Days	42,985	23,757	9,524	27,308	15,568	56,326
Non Distraction Days	43,868	20,928	9,430	32,613	16,509	60,960
Difference	-883	2,829	94	-5,305	-941	-4,634
p-val	0.245	0.999	0.625	0.000	0.021	0.019

Table VII

Asymmetric investor reaction to firm specific announcements during various distraction events

$$R_{it} = \beta_0 + \beta_{1i} R_{it-1} + \beta_{2i} R_{Mt} + \beta_{3i} D_t + \beta_{4i} Q_t + \varepsilon_{it},$$

$D_t = \{ D_{1t}, D_{2t}, D_{3t}, D_{4t} \}$ are dummy variables for Monday through Thursday,

$Q_t = \{ Q_{1t}, Q_{2t}, Q_{3t}, Q_{4t}, Q_{5t} \}$ are dummy variables for days for which previous 1 through 5 days are non-weekend holidays

$$\widehat{CAR[0,1]}_{it} = \gamma_0 + \gamma_1\, sent\_pos_{it} + \gamma_2\, sent\_neg_{it} + \gamma_3\, relevance_{it} + \gamma_4\, novelty_{it} + \gamma_5\, size_{it} + \gamma_6\, (P/B)_{it} + \gamma_7\, Retail\_pct_{it} + \gamma_8\, Retail\_pct_{it} * D_{pos} + \gamma_9\, Retail\_pct_{it} * D_{pos}\, (Industry\ Dummies)_i + (Year\ Dummies)_t + v_i$$

$\widehat{CAR[0,1]}_{it}$ are the cumulative abnormal returns over day 0 to +1; $sent\_pos_{it}$ and $sent\_neg_{it}$ are probability that the sentiment of the news was positive and negative respectively; $relevance_{it}$ measures the pertinence of the asset reported in the news; novelty is the measure of uniqueness of the news being reported; Retail_pct indicates the percent holdings by retail investors respectively

	All Distraction Days		**Natural Calamities & Disasters**		**Political**		**Law & Order**		**Sports & Entertainment**	
	Coef.	Robust Std Error	Coef.	Robust Std Error	Coef.	Robust Std Error	Coef.	Robust Std Error	Coef.	Robust Std Error
Sent_pos	0.546	0.147***	0.743	0.303**	-0.117	0.263	0.589	0.311*	0.824	0.288***
Sent_neg	-0.239	0.147	-0.227	0.312	-0.622	0.262*	0.315	0.308	0.088	0.282
Relevance	0.061	0.097	-0.077	0.203	-0.164	0.174	0.194	0.202	0.203	0.189
Novelty	0.001	0.012	-0.003	0.028	0.042	0.020*	-0.021	0.024	-0.035	0.021*
Size	-0.032	0.011***	-0.099	0.025***	-0.067	0.020***	0.009	0.024	0.012	0.022
P/B	0.004	0.003	0.020	0.009**	0.011	0.004*	0.005	0.003	-0.012	0.003***
Retail_pct	0.003	0.002*	0.001	0.006	-0.005	0.005	0.004	0.006	0.009	0.005*
Retail_pct*D_{pos}	0.001	0.004	-0.003	0.008	0.011	0.007	0.004	0.007	-0.011	0.007*
Retail_pct*D_{neg}	0.008	0.004*	0.009	0.009	0.016	0.008*	-0.011	0.009	0.008	0.008
Intercept	0.070	0.306	0.645	0.638	0.450	0.512	-1.625	0.902*	-0.345	0.614
Industry Dummies	Yes		Yes		Yes		Yes		Yes	
Year Dummies	Yes		Yes		Yes		Yes		Yes	
N	25,117		6,298		7,202		5,240		7,072	

Notes: Industries are defined by the Fama-French 48-industry classification. Variables are winsorized at the 1 percent and 99 percent levels. *,**, and *** indicate significance at the p < 0.10, p < 0.05, and p < 0.01 levels respectively

6. Conclusion

The main findings of the study highlight that investors underreact to both positive and negative sentiment during distraction periods. We point out that investors do not behave homogeneously in all categories of distraction. The underreaction is more pronounced during distraction periods triggered by natural calamities & disasters and political events compared to distraction caused by other events. We also examine the trading pattern of investors during these periods. Consistent with our hypothesis, we find a statistically significant drop in the number of transactions undertaken by market participants. This decline in both the turnover and traded quantity indicates that investor participation drops significantly during distraction periods. Further our results suggest that retail traders are more distracted by external distraction as compared to institutional investors. This underreaction holds for both positive and negative sentiment news. Minor underreaction by institutional investors could be due to the adoption of more sophisticated quantitative trading strategies and increased reliance on technology. Our results have possible managerial implications regarding timing the release of voluntary disclosures to manage investor expectations.

References

Bagnoli, M. E., Clement, M. B., & Watts, S. G. (2005). Around-the-Clock Media Coverage and the Timing of Earnings Announcements. *SSRN Electronic Journal*, (May)

Baker, M., & Wurgler, J. (2006). Investor Sentiment and the Cross-Section of Stock Returns. *The Journal of Finance*, *61*(4), 1645–1680

Baker, M., & Wurgler, J. (2007). *Investor Sentiment in the Stock Market*. *The Journal of Economic Perspectives* (Vol. 21). Cambridge, MA. Retrieved from http://www.nber.org/papers/w13189.pdf

Barber, B. M., & Odean, T. (2008). All That Glitters: The Effect of Attention and News on the Buying Behavior of Individual and Institutional Investors. *Review of Financial Studies*, *21*(2), 785–818

Barberis, N., Shleifer, A., & Vishny, R. (1998). A model of investor sentiment. *Journal of Financial Economics*, *49*(3), 307–343

Barman, P. C., Iqbal, N., & Lee, S.-Y. (2006). Non-negative Matrix Factorization Based Text Mining: Feature Extraction and Classification. In I. King, J. Wang, L.-W. Chan, & D. Wang (Eds.), *Neural Information Processing: 13th International Conference, ICONIP 2006, Hong Kong, China, October 3-6, 2006. Proceedings, Part II* (pp. 703–712). Berlin, Heidelberg: Springer Berlin Heidelberg

Blei, D. M. (2012). Probabilistic topic models. *Communications of the ACM*, *55*(4), 77

Boulland, R., Degeorge, F., & Ginglinger, E. (2016). News Dissemination and Investor Attention. *Review of Finance*, (May 2016), rfw018

Chakrabarty, B., & Moulton, P. C. (2012). Earnings announcements and attention constraints: The role of market design. *Journal of Accounting and Economics*, *53*(3), 612–634

Chakrabarty, B., Moulton, P. C., & Wang, X. (2015). Attention effects in a high-frequency world [Electronic version]. Retrieved [14-Jul-2017], from Cornell University, SHA School site: http://scholarship.sha.cornell.edu/workingpapers/19

Chen, J., Liu, Y.-J., Lu, L., & Tang, Y. (2016). Investor Attention and Macroeconomic News Announcements: Evidence from Stock Index Futures. *Journal of Futures Markets*, *36*(3), 240–266

Da, Z., Engelberg, J., & Gao, P. (2011). In Search of Attention. *The Journal of Finance*, *66*(5), 1461–1499

Daniel, K. D., Hirshleifer, D., & Subrahmanyam, A. (1998). Investor Psychology and Security Market Under- and Overreactions. *Journal of Finance, 53*(6), 1839–1886

De Long, J. B., Shleifer, A., Summers, L. H., & Waldmann, R. J. (1990). Noise Trader Risk in Financial Markets. *Journal of Political Economy*, *98*(4), 703–738

Dellavigna, S., & Pollet, J. M. (2009). Investor Inattention and Friday Earnings Announcements. *The Journal of Finance*, *64*(2), 709–749

Drake, M. S., Gee, K. H., & Thornock, J. R. (2016). March Market Madness: The Impact of Value-Irrelevant Events on the Market Pricing of Earnings News. *Contemporary Accounting Research*, *33*(1), 172–203

Edmans, A., Garcia, D., & Norli, Ø. (2007). Sports Sentiment and Stock Returns. *The Journal of Finance, 62*(4), 1967–1998

Egeth, H., & Kahneman, D. (1975). Attention and Effort. *The American Journal of Psychology*, *88*(2), 339

Feng, X., Jiao, Y., Lv, C., & Zhou, D. (2016). Label consistent semi-supervised non-negative matrix factorization for maintenance activities identification. *Engineering Applications of Artificial Intelligence*, *52*, 161–167

Francis, J., Pagach, D., & Stephan, J. (1992). The Stock Market Response to Earnings Announcements Released during Trading versus Nontrading Periods. *Journal of Accounting Research*, *30*(2), 165–184

Hirshleifer, D., Lim, S. S., & Teoh, S. H. (2009). Driven to Distraction: Extraneous Events and Underreaction to Earnings News. *The Journal of Finance, 64*(5), 2289–2325

Hirshleifer, D., & Teoh, S. H. (2003). Limited attention, information disclosure, and financial reporting. *Journal of Accounting and Economics*, *36*(1–3), 337–386

Hou, K., Xiong, W., & Peng, L. (2009). A Tale of Two Anomalies: The Implications of Investor Attention for Price and Earnings Momentum. *SSRN Electronic Journal*, 43

Lim, K. W., Buntine, W., Chen, C., & Du, L. (2016). Nonparametric Bayesian topic modelling with the hierarchical Pitman–Yor processes. *International Journal of Approximate Reasoning*, *78*, 172–191

Liu, B. (2012). Sentiment Analysis and Opinion Mining. *Synthesis Lectures on Human Language Technologies*, *5*(1), 1–167

Loh, R. K. (2010). Investor Inattention and the Underreaction to Stock Recommendations. *Financial Management*, *39*(3), 1223–1252

Manning CD, Surdeanu M, Bauer J, Finkel J, Bethard SJ, M. D. (2014). The Stanford CoreNLP natural language processing toolkit. In *Proceedings of 52nd Annual Meeting of the Association for Computational Linguistics : system demonstrations* (pp. 55–60). Baltimore, Maryland, USA.

Miller, E. M. (1977). Risk, Uncertainty, and Divergence of Opinion, *XXXII*(4), 1151–1169.

Odean, T. (1999). Do Investors Trade Too Much? *The American Economic Review*, *89*(5), 1279–1298

Pang, B., & Lee, L. (2008). Opinion Mining and Sentiment Analysis. *Foundations and Trends® in Information Retrieval*, *2*(1–2), 1–135

Shleifer, A., & Vishny, R. W. (1997). The Limits of Arbitrage. *The Journal of Finance*, *52*(1), 35–55

Xue, Y., Tong, C. S., & Yuan, J. Y. (2014). LDA-based Non-negative Matrix Factorization for Supervised Face Recognition. *Journal of Software*, *9*(5), 1294–1301

Yuan, Y. (2015). Market-wide attention, trading, and stock returns. *Journal of Financial Economics*, *116*(3), 548–564

Zhang, J., Zhang, L., Yang, G., Wu, D., Jiang, L., Huang, L., … Li, M. (2013). Nonnegative matrix factorization for the improvement in sensitivity of discovering potentially disease-related genes. *Chemometrics and Intelligent Laboratory Systems*, *126*, 100–107